MINN2020A-Report, Group 16

Aphelele Xasibe, Onkagile Moraka, Esihle Dwesini, Sithembiso Dlamini

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1 Abstract

This project focuses on the development of a Python-based safety and risk-alert program for a mining environment. The purpose of the program is to evaluate selected safety and operational indicators, including equipment temperature, vibration, PPE compliance, worker fatigue, incident severity, downtime, and overdue maintenance. In python conditional statements and predefined thresholds were used to identify potentially unsafe conditions and generate appropriate alerts. The project demonstrates the application of programming logic to mining-related problem-solving and decision support. The program is intended to assist users in recognizing conditions that may require attention. Its effectiveness is evaluated by testing the program with different input values and checking whether the alerts generated correspond to the specified conditions.

2 Introduction

Mining plays a significant role in the South African economy through the creation of jobs, industrial development, and the supply of valuable mineral resources. However, mining operations involve occupational health and safety risks, as well as the use of complex equipment that requires regular monitoring and maintenance. Poor monitoring of worker safety, equipment condition, and maintenance activities can contribute to unsafe working conditions, equipment failure, operational delays, and increased downtime.

The aim of this project is to design and develop a Mining Health, Safety and Equipment Monitoring Application using Python. The application involves mining data management, basic risk assessment, equipment condition monitoring, maintenance tracking, automated alerts, and data visualization within a user-friendly interface. It also aims to provide role-based access control so that different users can access functions relevant to their responsibilities. The purpose of the application is to help users identify potential safety risks, monitor incidents, detect abnormal equipment conditions, identify maintenance requirements, and assess equipment availability.

3 Background

In mining operations, maintenance is essential to improve the efficiency, performance, and quality of the processes involved in mineral extraction. Proper maintenance of mining equipment helps minimize total life-cycle costs, reduce long-term operating expenses, and, most importantly, improve workplace safety by preventing equipment failures that could create hazardous conditions, particularly in coal mines (Florea, Toderaş, and Itu, 2023). For all organizations and companies, a major challenge is encountering various production equipment failures, which ultimately lead to lost production time and result in increased operating costs. Therefore, establishing an effective maintenance strategy is essential to ensure reliable performance of production equipment. Many companies have adopted preventive maintenance, a proactive approach that involves regularly inspecting and maintaining equipment to prevent unexpected breakdowns (Florea, Toderaș and Itu, 2023). Effective monitoring of factors such as equipment temperature, vibration levels, worker fatigue, personal protective equipment (PPE) compliance, and overdue maintenance can provide important indications of potentially unsafe conditions. If these indicators are not identified and addressed appropriately, they may contribute to equipment failure, operational interruptions, incidents that threaten worker safety.

At Khwezi Mine monitoring several risk indicators can be challenging because each indicator may require different limits and responses. Capturing the data on manual inspections and paper- based records, makes the data hard to analyse and interpret, thus digital technology has enabled more faster and efficient ways of checking and analysing data. Computer programming provides a way to process input data systematically and identify conditions that require attention. Python is suitable for this purpose because it supports conditional statements, logical operators and functions that can evaluate values and generate outputs based on predefined criteria.

The project focuses on developing a Python-based mining safety and risk alert program. The program evaluates selected safety and operational indicators against specified conditions and generates alerts when thresholds are reached or exceeded. The aim is to support the identification of potential risks and demonstrate how programming can assist with safety monitoring and decision -making in a mining environment.

4 Methodology

The Khwezi Mining Health, Safety and Equipment Monitoring Application was developed using Python to provide a centralized system for managing mining safety, incidents, equipment conditions, and maintenance records. The development process led to using algorithms, flowcharts, pseudocode, python program and functional application as a simple approach to break down the complexity

of the project. The application uses the Tkinter library to create a graphical user interface and Pandas to read, process, update, and analyze data stored in CSV files. The data used and the file are simulated files using artificial intelligence.

The application generates demonstration datasets for workers, safety incidents, and mining equipment when there are no relevant CSV files. Users can add worker assessments and incident records, search and filter incidents, manage maintenance information, and export reports. Data validation is applied to selected inputs to identify missing information, duplicate worker IDs, and invalid numerical values. The development process involved creating separate modules for user authentication, worker safety assessments, incident management, equipment monitoring, maintenance tracking, risk assessment, and report generation. Role-based access control was implemented to assign permissions to administrators, safety officers, mining engineers, maintenance engineers, and managers. Passwords were also created for each position.

The risk assessment is performed by multiplying the likelihood and consequence ratings, each measured on a scale of 1 to 5. The resulting risk scores are classified as low, medium, high, or critical. Equipment condition monitoring uses predefined temperature and vibration thresholds to identify normal, warning, and critical conditions. Equipment availability is calculated by dividing operating hours by the sum of operating hours and downtime and then multiplying by 100.

A central dashboard presents key performance indicators, including PPE compliance, worker fatigue, safety incidents, equipment availability, and equipment condition. Visualizations show incident trends and equipment condition categories, while automated alerts identify potential safety concerns, abnormal equipment conditions, excessive downtime, and overdue maintenance. The application was designed to support data review, summary reporting, and CSV report export to assist mining personnel in monitoring operational and safety information.

5 Results

The application was run with its demonstration data (10 workers, 7 incident records, 8 equipment units); these records are placeholders to show behaviour, not real mine data. Table 5 summarises the indicators the system produced.

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Equipment condition. The drilling machine (EQ-004; 102°C, 9.1 mm/s) was classified Critical and the loader (EQ-002) and scraper winch (EQ-006) Warning, matching the thresholds in Table 1. The conveyor (EQ-008) has the lowest availability (65.5%) and the haul truck (EQ-001) the highest (95.1%).

Area	Indicator	Value
4*Workers	Total workers	10
	Average PPE compliance	91.2%
	Workers with PPE <80%	2
	High fatigue workers (≥8)	2
Incidents	Safety incidents / near misses / observations	2 / 2 / 3

Area	Indicator	Value
3*	High or critical severity	4
	High-risk observations	2
	Day vs. night shift records	4 vs. 3
4*Equipment	Normal / Warning / Critical	5 / 2 / 1
	Under maintenance	2
	Average availability	79.8%
	Units with downtime >20 h	4

Alerts. The dashboard listed the PPE, fatigue, high-risk observation, condition and downtime alerts above. An example for EQ-004 reads: CRITICAL · Equipment EQ-004: temperature 102°C, vibration 9.1 mm/s.

Business questions. The interface answers questions from Appendix A of the brief: incident counts and the department and shift with the most incidents (incident analysis and Reports tab), PPE compliance, near-miss counts, equipment with highest downtime or lowest availability, equipment requiring maintenance, average availability and the share of equipment operating normally (62.5% here). In the demonstration data the Production department records more incidents (4) than Engineering (3), but the sample is too small for statistical conclusions about shift or department differences.

Usability and robustness. Forms reject invalid numbers, dates and duplicate IDs with explanatory messages, actions give confirmation feedback, and navigation is limited to sections the role may use. Tables scroll and pages resize with the window.

6 Conclusion and Recommendations

6.1 Conclusion

This project focuses on developing a Mining Health, Safety, and Equipment Monitoring Application using Python to improve safety and equipment monitoring in mining operations. The application uses conditional statements and predefined threshold levels to detect potential risks related to equipment temperature and vibration, personal protective equipment (PPE) compliance, employee

fatigue, accident severity, equipment downtime, and overdue maintenance.

Different input values are entered into the application to test whether it generates the correct alerts according to the conditions that have been set. Overall, the project demonstrates how Python programming can be used to solve practical problems in the mining industry and support decision-making. However, the application is intended to assist with safety monitoring and equipment management rather than replace professional judgement, regular inspections, or compliance with established mine safety standards.

6.2 Recommendations

Further testing and improvements to the data validation process are recommended to enhance the application's performance and reliability. Testing the program under various conditions would help evaluate its accuracy, functionality, and ability to handle unexpected inputs. This would also help identify programming errors and determine whether the application generates appropriate alerts when specific conditions are met. Furthermore, strengthening data validation would help prevent inaccurate, incomplete, invalid, or unrealistic information from being processed. These improvements would ensure that the application produces more reliable results and provides useful information to support mining safety monitoring, equipment management, and informed operational decision-making.

7 References

Florea, V.A., Toderaș, M. and Itu, R.B., 2023. Assessment possibilities of the quality of mining equipment and of the parts subjected to intense wear. Applied Sciences, 13(6), p.3740.